KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 7 REPORT

Dimensionality Reduction

Team 16: Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team Member Name	Roll Number
Atchaya S	25BAD016

1. Week Contribution:

Interpreted relationships among personality features and the PCA results, and maintained the Week 7 notebook and project documentation in GitHub.

2. Work Done:

During Week 7, my individual work focused on interpreted relationships among personality features and the PCA results, and maintained the Week 7 notebook and project documentation in GitHub.

- Reviewed the correlation matrix for the seven personality features: Nscore, Escore, Oscore, Ascore, Cscore, Impulsive and SS.
- Identified the strongest observed relationship between Impulsive and SS, with a correlation coefficient of approximately 0.623.
- Reviewed the project's multicollinearity check and noted that no pair of personality features crossed the 0.80 correlation threshold used in the analysis.
- Studied the PCA output, where the seven personality features were represented using six principal components while retaining approximately 95.07% of the variance.
- Maintained the Week 7 dimensionality-reduction notebook and supporting documentation in GitHub.

3. Topics Covered:

- Correlation matrix and interpretation of feature relationships.
- Multicollinearity checking.
- Principal Component Analysis (PCA).
- Explained variance and dimensionality reduction.
- GitHub updates for Week 7 analysis files.

4. Outcome of My Contribution:

The analysis helped me understand how correlation and PCA can be used to study relationships and reduce the number of dimensions while retaining most of the information. The Week 7 files were maintained in the project repository.

5. GitHub Repository:

Repository: https://github.com/atchayas-ai/DSV_Team_16_Youth_Smoking_Drug_Abuse

I helped manage and maintain the shared repository by keeping the relevant weekly documentation, notebooks, datasets and project materials organised. The exact files updated during each week are described in the Work Done section.

6. Reference:

Project repository: GitHub – Team 16, Youth Smoking and Drug Consumption Analysis.